MIL-S-11031B
5 JUNE 1963
SUPERSEDING
MIL-S-11031A(ORD)
7 SEPTEMBER 1951

MILITARY SPECIFICATION

SEALING COMPOUND, ADHESIVE: CURING (POLYSULFIDE BASE)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers on type of inert, curing-type, adhesive sealing compound suitable for use by injection in bonding metal to metal or glass to metal, in optical instruments or fire control instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-116 — Tape, Paper, Gummed, Water-Resistant.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiberboard.

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

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MIL-A-101 — Adhesive, Water-Resistant for Sealing Fiberboard boxes.

MIL-L-15016 — Lubricating Oil, General Purpose.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 175 — Adhesives: Methods of Testing.

FED. TEST METHOD STD. No. 601 — Rubber: Sampling and Testing.

FED. TEST METHOD STD. No. 791 — Lubricants; Liquid Fuels, and Related Products; Method of Testing.

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MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 8030

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MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The sealing compound shall be a two-part material consisting of a base (polysulfide) compound and an accelerator.

3.2 Color. The base compound shall be black in color.

3.3 Volatile content. The volatile content of the freshly prepared sealing compound shall not exceed 3 percent by weight when tested as specified in 4.5.1.

3.4 Water solubility. The water solubility of the cured sealing compound shall not exceed 6 percent by weight in distilled water when tested as specified in 4.5.2.

3.5 Low temperature flexibility. The cured sealing compound shall show no evidence of cracking or lack of adhesion when subjected to flexing at —40°F. (—40°C.) and tested as specified in 4.5.3.

3.6 Oil resistance. The cured sealing compound shall show no evidence of blistering, softening, or loss of adhesion when in contact with petroleum oil and synthetic fluid and tested as specified in 4.5.4.

3.7 Adhesion. The cured sealing compound shall exhibit a bond strength of not less than 200 pounds per square inch when tested as specified in 4.5.5.

3.8 Effect of volatile components. There shall be no evidence of chemical attack, such as crazing, pitting, or etching on the surface of methyl methacrylate when exposed to volatile components of freshly cured sealing compound and tested as specified in 4.5.6.

3.9 Water vapor permeability. The cured sealing compound shall have a water vapor transmission rate of not greater than 0.025 grams per hour per square inch when tested as specified in 4.5.7.

3.10 Working life. The mixed sealing compound shall have a cone penetration value of not less than 200 after 3 hours at 75 ± 2°F. and the change in cone penetration within the 3 hour period shall be not greater than 75 when tested as specified in 4.5.8.

3.11 Hardness. The cured sealing compound shall have a Durometer A hardness of 40 ± 5 when tested as specified in 4.5.9.

3.12 Instructions for use. The supplier shall provide complete instructions for use of the sealing compound. The instructions shall either be on a label on the base compound, or on an instruction sheet in each shipment container of sealing compound and accelerator and shall contain the following information:

(a) Storage stability. Indicate the optimum storage temperature and conditions for the materials, and the expectant storage life of the materials under these conditions. Also date of manufacture.

(b) Mixing proportions. State the proportions of base compound to accelerator required to produce a sealing compound conforming to all tests specified in 4.5.

(c) Mixing techniques. Describe clearly the necessary techniques for mixing.

(d) Pertinent information. State any other pertinent information relative to storage, precautions, mixing or use of the sealing compound.

3.13 Workmanship. The workmanship shall be first-class in every respect. The com-

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switches.

pounding of the sealing compound shall be such that after thorough mixing of the base compound and the accelerator, the sealing compound shall be homogeneous.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 **Lot.** A lot shall consist of one batch or mix each of base compound and accelerator from one manufacturer and offered for inspection at one time.

4.3 **Sampling.**

4.3.1 *For examination.* A random sample of filled containers shall be selected for examination in accordance with level I of Standard MIL–STD–105.

4.3.2 *For tests.* A representative 1-quart sample of base compound and a corresponding quantity of hardener shall be taken from each lot. All of the tests shall be performed on this sample.

4.4 **Examination.** Sample units selected in accordance with 4.3.1 shall be examined for the defects and at the acceptable quality level shown in table I.

TABLE I. *Methods of examination*

Category	Material	AQL percent	Defect	Method of inspection
Critical	None defined			
Major 101	Unit containers (base compound and accelerator). See 4.3.1, (5.1.1.1 or 5.1.1.2) and 3.12.	2.5	Improper size	Visual.
Major 102			Wrong type	Visual.
Major 103			Improper fill[1]	Visual and scale.[2]
Major 104			Improper proportions	Visual.
Major 105			Leakage	Visual.
Major 106			Improper closure	Visual.
Major 107			Missing or improper instructions	Visual.
Major 108	Boxes interior (if required) (See 4.3.1 and 5.1.1.2).	2.5	Wrong type box	Visual.
Major 109			Wrong size	Visual.
Major 110			Not sealed properly	Visual.
Major 111	Exterior Box, (see 4.3.1 and 5.2 and 5.3).	2.5	Wrong type	Visual.
Major 112			Improper size	Visual.
Major 113			Lack of or improper strapping	Visual.
Major 114			Gross weight, max.	Approved scale.[2]
Major 115			Marking misleading or unidentifiable	Visual.
Major 116			Improperly closed	Visual.

[1] A container filled with the minimum required quantity of material shall be weighed, and this weight compared with the weight of unfilled containers to determine fill.

[2] Approved by procuring activity.

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4.5 Tests

4.5.1 *Volatile content.* Not less than 10 grams of the base compound and accelerator shall be prepared at one time. The ratio of the base compound to accelerator shall be as specified by the manufacturer (see 3.12). The two components shall be thoroughly mixed taking care that all traces of the accelerator disappear and that no base compound is left around the edges or in the corners of the mixing vessel. Examination shall be made for homogeneity (see 3.13). A sample (about 2 grams) of the prepared compound shall be spread uniformly, approximately 1/16 inch thick, on one side of a weighed 2 by 4 by 0.040 inch aluminum panel. Immediately after preparing the panel, it shall be accurately weighed and placed in a circulating air oven at 200 ± 2°F. (93.3° ± 1.1°C.) for a period of 2 hours. The panel shall then be removed, and allowed to cool to room temperature, then carefully reweighed. The loss in weight divided by the initial weight of the sample, multiplied by 100, shall constitute the percent of volatile content. The test shall be performed in duplicate. Failure of either sample shall be cause for rejection.

4.5.2 *Water solubility.* The panels with samples, which were used in determining the volatile content, shall be immersed in distilled water and heated at 160 ± 2°F. (71.1° ± 1.1°C.) for 4 hours. They shall then be removed and dried in air at 200 ± 2°F. (93.3° ± 1.1°C.) for 2 hours. The loss in weight divided by the weight of the sample after the determination of the volatile content and multiplied by 100 shall constitute the percent of water-soluble material.

4.5.3 *Low temperature flexibility.* A 1 by 4 by 0.040 inch aluminum panel shall be coated on one side with the freshly prepared compound (see 4.5.1) to a thickness of 1/16 inch. The panel shall be allowed to stand for 72 hours at 75 to 85°F. or placed in a circulating air oven for 2 hours at 200 ± 2°F. (93.3 ± 1.1°C.). The panel shall then be subjected to a temperature of —40 ± 2°F. (—40 ± 1.1°C.) for 4 hours. At that temperature, the panel shall be bent rapidly over a 1 inch diameter mandrel through an angle of 90° and with the uncoated surface resting on the mandrel. The mandrel shall be at the same temperature as that of the panel. After bending the panel, the sample shall be inspected for evidence of cracking and lack of adhesion. Test shall be performed in duplicate. Failure of either sample shall be cause for rejection.

4.5.4 *Oil resistance.* Approximately 1 gram of the freshly prepared compound (see 4.5.1) shall be spread uniformly on one side of a 1 by 4 by 0.040 inch aluminum panel. The sample shall be allowed to cure for 72 hours at 75 to 85°F. or for 2 hours at 200 ± 2°F. (93.3° ± 1.1°C.). The sample shall then be partially immersed for 4 days in mineral oil conforming to symbol 3050 of Specification MIL-L-15016. This test shall be conducted at room temperature. The panel shall be removed and the sample shall be examined for evidence of blistering, softening, and lack of adhesion. Test shall be performed in duplicate. Failure of either sample shall be cause for rejection.

4.5.5 *Adhesion.* The freshly prepared compound (see 4.5.1) shall be applied to both surfaces of a steel tensile specimen as described in method 1011.1 of Fed. Test Method Std. No. 175. The specimen shall be compressed under a load of 200 pounds for about 20 seconds to insure a minimum gap between the adherends of the specimen. The test specimen shall then be conditioned for 72 hours at 75 to 85°F. Bond strength of the assembly shall be measured in a tensile testing machine whose idle jaw separation speed is 1 inch per minute. The test shall be conducted at room temperature. Test shall be performed in duplicate. Failure of either specimen shall be cause for rejection.

4.5.6 *Effect of volatile components.* A 5 to

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6 gram sample of freshly prepared compound (see 4.5.1) shall be placed on a 2 inch diameter watch glass and then placed in an inclosed vessel which contains a polished methyl methacrylate window as shown on figure 1. The test fixture shall be placed in an oven at 160 ± 2°F. (71.1° ± 1.1°C.) for 3 hours, then removed and allowed to cool to a temperature of approximately 100°F. (38°C.). The outer surface of the methyl methacrylate window shall be cooled by placing a piece of dry ice having a flat side of about ½ square inch onto the window of the fixture and allowed to remain in contact with the window until condensation is observed on the inner surface of the window or for a period not greater than 20 minutes. The fixture shall then be allowed to remain at room temperature for a period of 2 hours. The window of the fixture shall be removed and its inner surface washed with tap water and wiped with a soft cloth. The surface shall then be examined with the aid of a 10 power magnifier (approximately) in the area where condensation had formed for visual evidence of crazing, piting, or etching. Test shall be performed in duplicate. Failure of either sample shall be cause of rejection.

4.5.7 *Water vapor permeability.* A sufficient amount of freshly prepared sealing compound (see 4.5.1) shall be pressed between two polyethylene sheets to form a film about 0.075 inch thick and 3 inches in diameter. The sandwich thus formed shall be placed on dry ice for a period of about one hour. At this low temperature, one of the polyethylene sheets shall be peeled off and the sealing compound then allowed to return to room temperature. A suitable size screen which will retain the sealing compound shall be placed on the sealing compound. It is recommended that this screen be brass or aluminum of approximately 2.5 inches in diameter, having a mesh of approximately 33 by 40 and a wire thickness of 0.010 inch. The polyethylene sheet which was removed after freezing shall be placed on the screen and the assembly shall be compressed until the overall thickness of the sealing compound and screen will be 0.064 ± 0.002 inch. The assembly shall then be placed in an oven at 200 ± 2°F. (93.3° ± 1.1°C.) for 2 hours to cure and solidify the sealing compound. After the curing treatment, the polyethylene sheets shall be removed and the screen-supported film shall be trimmed and placed in a position as shown on figure 2. Approximately 5 milliliters of distilled water shall be placed in a Petri dish. The two metal plates holding the film of sealing compound and the Petri dish shall be screwed together tightly to insure a good seal between the edge of the Petri dish and the sealing compound.

4.5.7.1 The two assemblies shall be prepared as described in 4.5.7. A third assembly shall be prepared in the same manner except that no water shall be placed in the Petri dish. The three assemblies shall be carefully weighed and placed in an oven at 160 ± 2°F. (71.1° ± 1.1°C.) for 48 hours. The loss in weight of the dry assembly shall be subtracted from the weight of each of the other two assemblies and the water vapor transmission rate shall be calculated in units of grams of water per hour per square inch of open area. The open area is the area of the 1.875 inch hole in the metal plate minus the area occupied by the wires of the screen within the hole. Failure of either sample shall be cause for rejection.

4.5.8 *Working life.* Prepare a sample of 75 to 80 grams of freshly prepared sealing compound (see 4.5.1). The sample shall be placed immediately into a receptacle of approximately 2 inches in diameter and 1¼ inch deep. The time shall be recorded and cone penetration test shall be made immediately thereafter. The test shall be made at a temperature of 75 ± 2°F. and in accordance with method 311.5 of Fed. Test Method Std. No 791. After 3 hours, the same penetration shall be repeated. The test shall be performed in duplicate. Failure of either sample shall be cause for rejection.

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4.5.9 *Hardness.* The test procedure shall be as described in method 3021 of Fed. Test Method Std. No. 601.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level A or C as specified (see 6.1).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* The base compound shall be packaged in 1-pint or 1 quart multiple-friction-top containers conforming to type IV of Specification PPP-C-96. The accelerator shall be packaged separately in suitable containers with sufficient quantity of accelerator in each container for use with each container of base compound.

5.1.1.2 *Intermediate packaging.* One container of the compound and one container of accelerator shall be immobilized in a fiberboard box conforming to Specification PPP-B-636. All seams of the box shall be sealed with water-resistant tape conforming to Specification UU-T-116.

5.1.2 *Level C.* Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B or C as specified (see 6.1).

5.2.1 *Level A.* Intermediate packages shall be packed in overseas class or type wood-cleated fiberboard boxes, wood-cleated-plywood boxes, nailed and lock-corner wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes conforming to Specifications PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-585 style 2, 2A or 3 or PPP-B-636 class 2 respectively. The gross weight of the shipping container shall not exceed 70 pounds. All flaps of corrugated and solid fiberboard boxes shall be securely sealed with water resistant adhesive applied to the entire area of contact between the flaps, or by a combination of stitching one set of flaps prior to packing and sealing the other flaps with water resistant adhesive. Water-resistant adhesive shall conform to Specification MIL-A-101.

5.2.2 *Level B.* Intermediate packages shall be packed in domestic class or type shipping containers conforming to the requirements of Specification PPP-B-591, PPP-B-601, PPP-B-585, Class 2, 2A or 3, PPP-B-621 or PPP-B-636 class 1. The gross weight of shipping containers shall be limited to approximately 200 pounds. As far as practical containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required and contain identical quantities. Containers shall be closed and strapped in accordance with applicable container specification or appendix thereto.

5.2.3 *Level C.* The material shall be packed in accordance with manufacturer's standard practice in containers of the type and size commonly used for the purpose in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.** In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size of container required.

(c) Level of packaging and packing required (see section 5).

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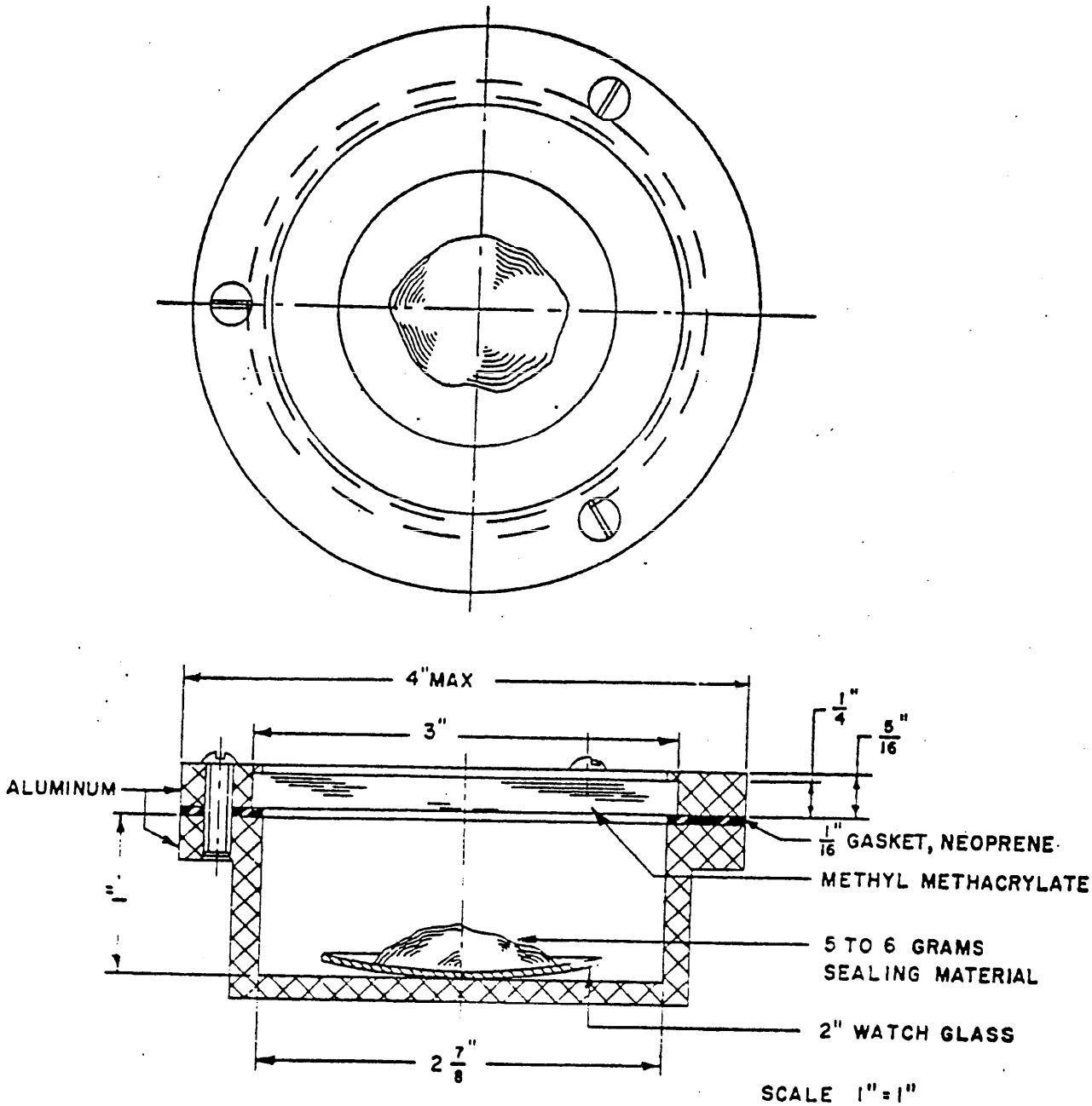
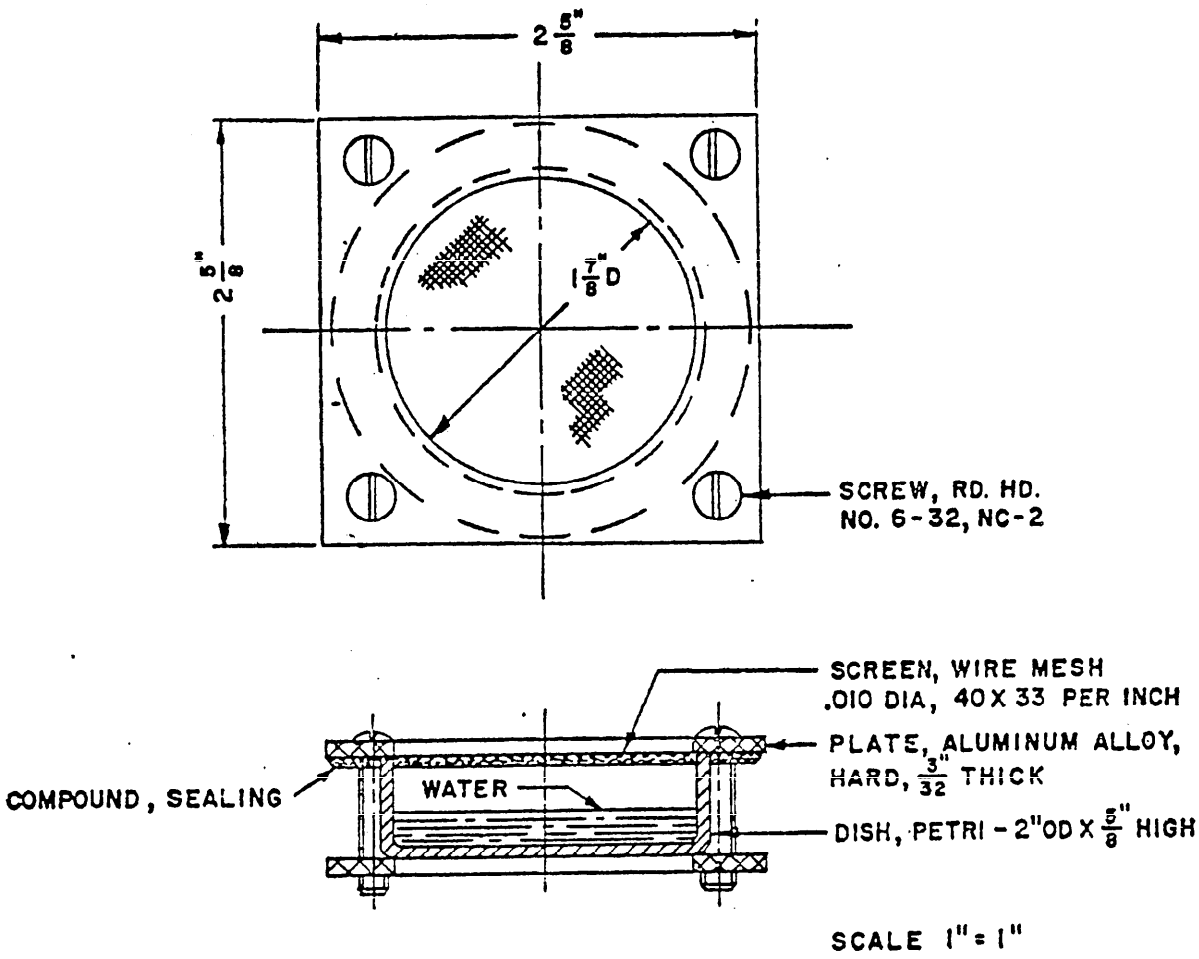


FIGURE 1. *Assembly for test on effect of volatile components.*

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NOTE:

TOP & BOTTOM PLATES ARE IDENTICAL, EXCEPT THAT SCREW HOLES ARE TAPPED INTO BOTTOM PLATE.

FIGURE 2. *Assembly for water vapor permeability test.*

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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